10 Parkinson's Disease

[T]he unhappy sufferer has considered it as an evil, from the domination of which he had no prospect of escape.
—James Parkinson, 1817 [Par17]

10.1 Overview

Parkinson's disease was first described by James Parkinson in a monograph published in 1817 [Par17]. It is a degenerative neurological condition, and we attempt to condense its neurological manifestations into four key signs: tremor, rigidity, bradykinesia (slowness of movement), and postural instability. But a far better picture of the signs of this condition was captured in William Gowers's short description in his 1901 textbook [Gow01], which flanked his famous sketch of a typical patient walking (figure 10.1):

> [T]he aspect of the patient is very characteristic. The head is bent forward, and the expression of the face is anxious and fixed, unchanged by any play of emotion. The arms are slightly flexed at all joints from muscular rigidity, and (the hands especially) are in constant rhythmical movement, which continues when the limbs are at rest so far as the will is concerned. The tremor is usually more marked on one side than on the other. Voluntary movements are performed slowly and with little power. The patient often walks with short quick steps, leaning forward as if about to run. [Gow01]

We have learned a considerable amount about the neurobiology of Parkinson's disease since the nineteenth century, but we do not know how to prevent it, and all of our present-day treatments remain palliative [LHR09]. The discovery by Carlsson and colleagues that a precursor to dopamine in the brain could ameliorate the effects of dopamine depletion [CLM57] led to successful medical therapy of Parkinson's disease with L-DOPA. Early on in the disease, L-DOPA provides the chemical precursor to produce more of the waning dopamine neurotransmitter, but gradually, fewer of these cells remain that can benefit from such a boost in chemical processing. Furthermore, there are two side effects that patients find especially disturbing: (1) the returning symptoms during the wearing-off phase of the drug, producing more radical on-off swings in motor symptoms, and (2) the gradual development

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Figure 10.1
A *well-marked case of this disease*, as described by William Gowers. Reproduced from the second edition of his 1901 textbook [Gow01].

of involuntary movements termed *dyskinesias* [SEJP09]. Although drug therapy remains the firstline standard of treatment for patients with Parkinson's disease, the long-term medical side effects of phamacological therapy has kept alive the surgical treatment options for phamacologically intractable Parkinson's disease.

There are three surgical lesion targets in the brain that have been found to reduce the symptomatology of Parkinson's disease effectively: the ventral intermediate nucleus (VIM) of the thalamus, the internal segment of the globus pallidus (GPi), and the subthalamic nucleus (STN). Because patients with Parkinson's disease generally require treatment on both sides of the brain, the efficacy of single-sided lesion treatment was often tempered by the complications of bilateral therapy. One never wants to lesion a brain symmetrically. So lesions can be administered asymmetrically. One also is hesitant to use too large a lesion at any one time, so there was a frequency of having to return to surgery to enlarge a lesion that was not effective enough.

Parkinson's disease is now the disease most widely treated by deep brain stimulation (DBS) [KJOA07]. In 1997, the U.S. Food and Drug Administration approved its use in Parkinson's disease. DBS is effective when used bilaterally, without the symmetrical lesion complications. Stimulation of the same targets that were lesioned produced palliative effects [KPLA99]. VIM lesions and stimulation were well characterized to reduce tremor,

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but although tremor is a hallmark of the disease, it is not the most disabling symptom for most patients. To better deal with the bradykinesia and rigidity, the GPi and STN became the preferred targets for stimulation. Although STN is now the dominant DBS target, it is unclear whether GPi, with a less burdensome set of cognitive side effects, might be better for some patients [ABH+05]. As our experience with DBS has progressed, a direct comparison of pharmacological versus DBS (bilateral GPi or STN stimulation) for Parkinson's disease found DBS to have advantages in quality of life outcomes despite the risks inherent in surgical treatment [DSBK+06, WFS+09]. Nevertheless, interest remains in lesions, whose effectiveness versus pharmacological therapy has also been shown [VBF+03], and whose long-term medical management and costs are considerably less than for patients who require life-long maintenance of DBS systems [BH06]. An especially compelling study of bilateral subthalamotomy argues the rationale for lesions and demonstrates the apparent safety and risk assessment of symmetric subthalamic nucleus lesions [AML+05]. The issue requires continued debate, as the majority of patients with Parkinson's disease on the planet, and their health care systems, lack the resources for DBS therapy.

The lesion debate notwithstanding, there is no more fruitful arena to consider a radical new approach to neural systems control than DBS technologies for Parkinson's disease. Our present approach has been to focus on high-frequency stimulation (130 Hz) delivered in open loop without feedback sensing. Along with the rise of such empirical DBS therapy, we have developed increasingly sophisticated computational models of the fundamental networks involved in the pathophysiology of Parkinson's disease. And we have developed increasingly sophisticated models of the physics of DBS stimulation to help us understand how DBS interacts with neurons. Let's examine this in detail.

10.2 The Networks of Parkinson's Disease

A great deal of the brain, especially the regions beneath the cortex, is heavily involved with movement regulation. Such areas include the connected set of basal ganglia, portions of the thalamus, and the cerebellum. The cerebellum may well contain as many neurons as the rest of the brain. Coordination and movement require a lot of mind.

In Parkinson's disease, there is degeneration of neurons that use dopamine as a neurotransmitter, which have their cell bodies in the substantia nigra at the upper edge of the midbrain. The decrease in neural output from the substantia nigra causes a disturbance in the network balance of excitation and inhibition, as schematized in figure 10.2. The result is a net increase in inhibition from GPi to thalamus (for a much more detailed discussion of the circuitry, see [OMRO+08]). But the lines and arrows in these static diagrams refer to average firing rate or activity and do not reflect the dynamics that is critical to understand what is happening. In Parkinson's disease the inhibition to the thalamus becomes phasic and oscillates.

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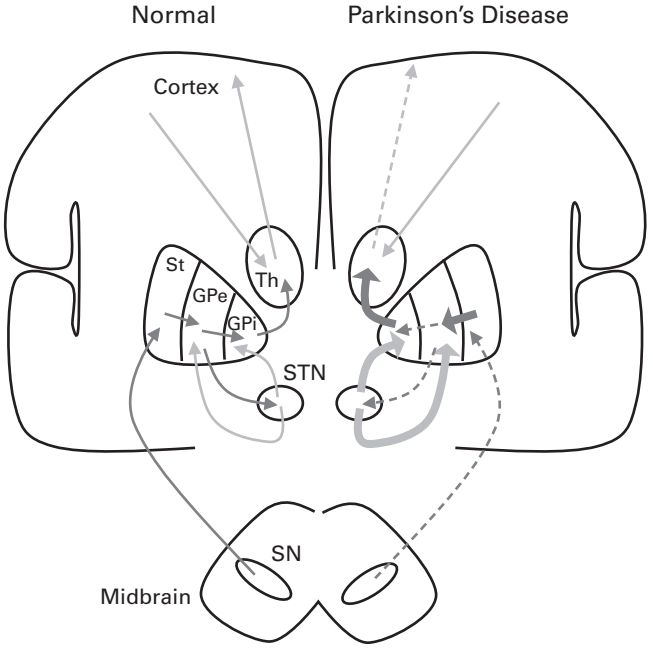


Figure 10.2 (plate 22)
An extremely simplified schematic of network imbalance in Parkinson's disease. Excitation is shown in red, and inhibition in blue. The normal brain (left) is contrasted with the brain in the Parkinson's disease state on the right, where thickened (thinned) lines indicate an increase (decrease) in excitation (red) or inhibition (blue). St, striatum; GPe, globus pallidus externa; GPi, globus pallidus interna; Th, thalamus, STN, subthalamic nucleus; SN, substantia nigra. I have made no distinction between indirect and direct pathways, and customized this for the purposes of the discussion within this chapter. For a more complete and detailed description of this anatomy, see [OMRO+08].

If, as in figure 10.2 (plate 22), you place a thin probe and burn a small hole in the GPi or STN, the symptoms of tremor, bradykinesia, and rigidity decrease.[1] We have understood this improvement in symptoms as resulting from a decrease in the excessive inhibition streaming out of the GPi and into the thalamus. If, instead, you place a thin electrode into these same structures, and stimulate at 130 Hz, you will get an almost identical decrease in the symptoms of tremor, bradykinesia, and rigidity. And this paradox requires explanation.

10.3 The Thalamus—It's Not a Simple Relay Anymore

The thalamus (Latin for *inner chamber*) is a walnut-sized piece of very high-end real estate in the center of the brain. We used to view it predominantly as a relay center for things

1. This is done by image- and microelectrode-guided stereotactic placement of a thin probe to the target nucleus within the brain, and the use of a calibrated radiofrequency thermal lesion.

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such as sensory information. The thalamus has a variety of relay stations resynapsing touch, vibration, sound, and visual information onto a fresh set of neurons coursing to their respective cortical targets (see schematic in [GS02]). With substantial damage to the thalamus such as that caused by a stroke, the brain loss consciousness [PP07]. Neuroscientists, evolutionary biologists, epileptologists, and philosophers have always gravitated to the human cortex as the object of their fascination. But gradually the reality of a cortex heavily interconnected with a series of relay loops to the deep thalamus has emerged, substantially complicating earlier views [GS02].

Transmission within the cortex tends to be slow, with local conductance velocities in the range of centimeters per second, in contrast to the fast long-range neural connections where many tens of meters per second are commonly observed. We think that the thalamus forms an essential transcortical relay system to help integrate information processing across cortical areas where local conduction speeds would otherwise render us far dimmer.

The reliability of this relaying of motor information through the thalamus is a focus of the following computational theory. Nevertheless, one must ask why, if there is such an intimate relationship between cortex and thalamus, does so much of the medical treatment focus on the lesioning or stimulation of targets many inches beneath the surface of the brain? Recent clinical trials have explored this issue, but so far, the results of these early efforts at superficial cortical stimulation have not been as effective as DBS or lesions [GSAV$^+$09].

10.4 The Contribution of China White

The study of the dynamics of Parkinson's disease was immeasurably helped by the seemingly inherent cravings of humans for narcotics. In the late 1970s synthesizing heroin-like compounds such as MPPP,[2] perhaps with a bit too much heat or acid in the reaction, produced a byproduct of the related MPTP.[3] A solitary report of poisoning from this byproduct quietly appeared in 1979 [DWM$^+$79]. By 1982, a clandestine drug chemist in California was selling this product under the name *china white*, until a bad batch started producing profound Parkinson's disease symptoms in a group of young addicts [LBTI83]. It was quickly discovered that this compound could produce the same symptoms in nonhuman primates [BCM$^+$83]. Both the human and animal data showed that MPTP was very selective in destroying the dopamine-containing neurons in the pars compacta of the substantia nigra—the same substructure that prominently degenerated in human Parkinson's disease. The symptoms and signs of the drug-induced and natural disease were nearly identical, and both diseases responded to treatment with the dopamine precursor drug L-DOPA.[4] The full story of these events is remarkable [LP95, FW02].

2. 1-methyl-4-phenyl-4-proprionoxy-piperidine.
3. 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine.
4. L-3,4-dihydroxyphenylalanine.

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The critical benefit from these events was the proof that loss of a small group of neurons in the substantia nigra could produce the triad of tremor, rigidity, and bradykinesia. The primate animal model provided us with a way to dramatically increase both our knowledge of the electrophysiology of the neuronal networks involved, and their potential electrical modulation, to the extent that the following can be described.

10.5 Dynamics of Parkinson's Networks

Prior to 2002, most models of Parkinson's disease were displayed as static diagrams (as in figure 10.2). Nevertheless, the advent of the MPTP primate model [WD03, RFMF$^+$01], and increasingly the recording of neurons from human Parkinson's patients during deep brain surgery [MMJ00, BOM$^+$01], revealed that the neurons within the Parkinsonian networks were strongly oscillatory (thalamus, GPi, and the external segment of the globus pallidus, GPe). In a fundamental experiment, Plenz and Kital [PK99] observed that if they mixed bits of GPe and STN in tissue culture, the cells would spontaneously connect and generate oscillations.

In the schematic of the basal ganglia, fundamental features of *central pattern generators* are seen in the connectivity of GPe and STN. GPe cells inhibit each other as well as STN. STN excites GPe. When STN cells are inhibited, they also demonstrate an exaggerated rebound excitation [BMT$^+$02]. Any simple neuronal membrane will spike if released suddenly from an inhibitory input—*anode break excitation* (chapter 3) [HH52d]. Some of the neurons critical to Parkinsonian dynamics have additional currents that exaggerate such inhibition-induced rebound excitation.

David Terman and colleagues [TRYW02] set out to explain these network effects on the basis of the biophysical properties of the individual neuronal types and their synaptic connectivity. They focused on the essence of what appeared to be the rhythm-generating circuitry, which turns out to also be the targets for both lesioning and stimulation in surgical therapy. (Their schematic is reproduced in figure 10.3.)

Applying typical experimental techniques in brain studies, we can record from single cells. But only in exceptional circumstances do we survey enough pairs of connected cells to develop strong characterizations about the details of the connectivity. There have been heroic experiments where investigators spent years carefully sticking microelectrodes into pairs of neurons to directly sample such connectivity [TM91], but such efforts remain rare. To make up for the paucity of connectivity data, Terman and colleagues [TRYW02] explored a range of connectivities. They had a great deal of intracellular data from networks from brain slice experiments that preserved native network architecture and connectivity to draw on (see [HB05] and references therein).

Three types of network topologies were constructed by Terman and colleagues [TRYW02]: random and sparsely connected, structured and sparsely connected, and structured and tightly connected, as shown in figure 10.4.

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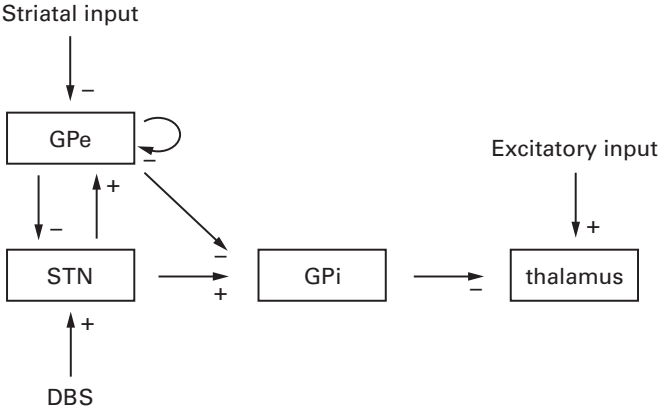


Figure 10.3
Schematic of rhythm-generating structures in Parkinson's disease. Striatal input refers to the outer segments of the basal ganglia that send input to these deeper segments. Excitatory input refers to sensorimotor input to the thalamus that needs to be relayed to cortex. Excitation is indicated by +, and inhibition by −. Reproduced from [RT04] with permission.

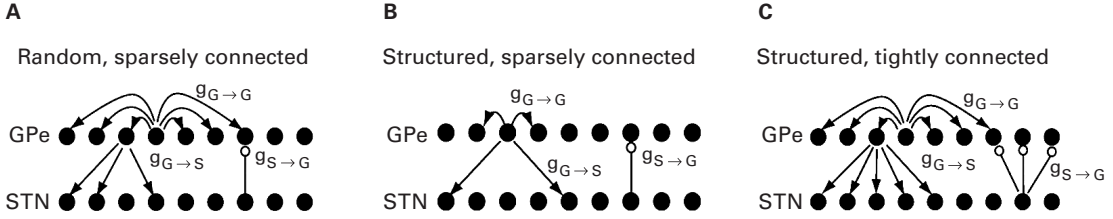


Figure 10.4
(A) Sparse random network. Each STN neuron excites a single random GPe cell, and each GPe neuron inhibits three random STN cells. GPe cells inhibit each other through all-to-all coupling. (B) Sparse structured network. Although more structured than the random sparse network of A, it is designed to avoid direct reciprocal connections between STN and GPe cells. Each STN neuron excites the single closest GPe cell, and each GPe neuron inhibits two STN cells skipping the three closest. GPe cells inhibit two immediate neighboring GPe cells. (C) Tightly connected structured network. Each STN neuron excites three closest GPe cells, and each GPe neuron inhibits the five closest STN cells. GPe cells inhibit each other through all-to-all coupling. Spatially periodic boundary conditions are applied (the network wraps around on itself). Reproduced from [TRYW02] with permission.

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To model the STN neurons, and bring out their rebound excitability, Terman and colleagues [TRYW02] model the STN membrane current as a combination of

$$C_m \frac{dv}{dt} = I_L - I_K - I_{Na} - I_T - I_{Ca} - I_{AHP} - I_{G \to S} \tag{10.1}$$

where C_m is membrane capacitance.

The first three currents ($I_L - I_K - I_{Na}$) are the Hodgkin-Huxley leak, rectifying potassium, and fast sodium, which evolution elected to conserve from cephalopods to mammals [HH52d] (chapter 3). In this case, these currents had parameters tailored to the mammalian neurons rather than the squid axon. $I_{G \to S}$ represents the current injected into these cells from the GPe synapses. And now three special currents will be described: I_{AHP}, I_{Ca}, and I_T.

The afterhyperpolarization current, I_{AHP}, is a potassium channel that opens in response to increasing amounts of intracellular calcium. While it is on, it prevents the cell from firing another spike, so it takes an important role in turning off excitation, as well as regulating firing frequency. It's a good way to create resonance frequency ranges within which a cell would prefer to fire. And it's a good way to generate a pacemaker. It is prominent in motoneurons in the spinal cord, where the frequency of discharge must be matched to the characteristics of the muscle fibers they connect to (fast or slow) [KBC99]. I_{AHP} is prominent in the suprachiasmatic nucleus where it helps translate the molecular clock of our circadian rhythms into neuronal firing frequency [CS03]. And I_{AHP} is important in the subthalamic nucleus, helping to make the neurons more sensitive to inputs at motor frequencies [BW99], and helping to create an oscillatory central pattern generator out of the network it is embedded within in the basal ganglia [BMT+02].

The high-threshold calcium current, I_{Ca}, is a representation of what are likely several high-threshold calcium channels in such cells [SBOM00]. High threshold means that they are activated with depolarization. Because the calcium Nernst reversal potential is very positive (even more positive than sodium), these channels in general support regenerative potentials (that is, they boost depolarization already in progress). Bursting cells tend to have such channels.

The low-threshold calcium current, I_T, is prominent in thalamic neurons where following inhibition such neurons rebound burst fire [MB97]. This current activates on hyperpolarization, and then deactivates more slowly than I_{Na} deactivates as the neuron depolarizes. Since the reversal potential for calcium is positive, activating such a current gives the neuron a boost to move its membrane potential away from rest and accentuate the tendency of any neuron to rebound a bit (as anode break excitation). I_T plays a role in the prominent postinhibitory rebound spiking seen in STN cells [BMT+02]. Interestingly, the same T-type calcium currents play a role in the automaticity and pacemaker function in heart cells, and can play a role in certain cardiac arrhythmias [VTA06].

Terman and colleagues used the same currents as in equation (10.1) for modeling GPe neurons, but changed their proportions to better match the firing properties seen in a variety of experimental studies [TRYW02].

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In the sparse random network (figure 10.4A), the strength of connectivity (maximal synaptic conductance) was varied from STN to GPe, and within the GPe to GPe network. Increasing the STN to GPe coupling produced a range of behaviors from sparse irregular firing, to episodic bursting, to continuous firing. The episodic regime was qualitatively similar to that reported in the classic study by Mahlon DeLong for cells in the normal primate GPe [DeL71]. One of the counterintuitive features of increasing the GPe to GPe inhibition is that it can increase the spread of activity through rebound. In the modeling of Terman and colleagues [TRYW02], the episodic burst firing is terminated by I_{AHP} as calcium is built up in GPe cells during the high-frequency firing episodes.

In the sparse structured network (figure 10.4B), there was more temporal clustering in the interactions among these cells, reflecting an increase in the topological spatial clustering of the neuronal connections.

In the tightly connected structured network (figure 10.4C), they also wrapped the network around on itself (periodic boundary conditions). This network generated waves that traveled. Initiating such waves required symmetry breaking in the STN-to-GPe and GPe-to-STN footprints. To form a solitary traveling wave, the GPe-to-GPe inhibitory footprint had to be larger spatially than the STN-to-GPe footprint (supporting a Turing instability [CH93, Mur03]). Key here was that, for a wave to be propagated, the STN and GPe cells had to have structured footprints so as to orderly spread activity to cells ahead of the leading edge of the wave [TRYW02].

Traveling waves have been experimentally observed in thalamic slices generating what are called spindle oscillations [KBM95]. Spindle waves have been related to sleep physiology, but the relationship to Parkinson's disease is not presently clear. Nevertheless, whenever there is evidence of temporal oscillations in a neural network, it is worth considering what the spatial structure of those oscillations are [HTY$^+$04]. Especially when networks are sparsely connected (as opposed to all-to-all connected where there is no meaningful spatial structure), considering whether waves might underlie such rhythms is reasonable.

Strong oscillations emerge in the GPe-STN network in Parkinson's disease and in dopamine depletion in experimental animals. Nevertheless, a body of experimental evidence, in human patients as well as MPTP primates (see [TRYW02]), fails to find the sort of highly correlated and synchronized firing that would support the coherent waves predicted in the most structured networks of figure 10.4. The picture emerging from this work is that, among the more sparse networks, the conversion from normal to Parkinsonian dynamics fit well with the schematic in figure 10.5. This schematic illustrates that following a loss of dopamine input to the striatum, a strengthening of striatal input to the GPe, perhaps with a concurrent weakening of recurring inhibitory connections within the GPe, could create a Parkinsonian state.

Now go back to the standard firing rate model of Parkinson's disease from figure 10.2. In this static model, a *decrease* in inhibition to the striatum led to an *increased* activity of STN, which *increased* excitation to GPe and GPi, which *increased* the inhibition to the thalamus. No oscillations arise in the static model, in a disease whose hallmarks are the

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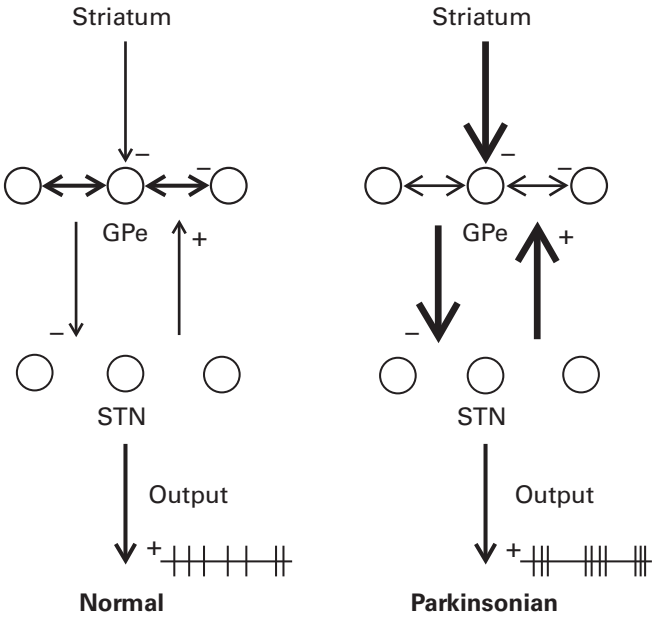


Figure 10.5
Schematic of Terman and colleagues [TRYW02] suggesting how an increase in striatal input and decrease in GPe internal connections would generate the oscillations of a Parkinsonian state. Reproduced with permission.

oscillations both within and outside the brain. In the view of Terman and colleagues, the level of striatal input (inhibitory) is a major regulator of whether oscillations arise from within this network that is naturally wired and poised to oscillate. Increased striatal inhibition effectively strengthened the coupling from GPe to STN. Decreasing intra-GPe inhibition would promote clustered activity within GPe. And inhibition, as is found in other dynamic phenomena in the nervous system such as seizures, plays multifaceted and complex roles [ZCBS06, MM08]. Inhibition in neuronal systems, similar to inhibitory reactants in reaction diffusion systems, is crucial in organizing patterns into clustered as opposed to homogeneous patterns [CH93, Mur03].

10.6 The Deep Brain Stimulation Paradox

Deep brain stimulation of the GPi or STN has almost identical effects on the symptoms of Parkinson's disease as do lesions of the GPi or STN. We have evidence of excessive activity of GPi in Parkinson's disease, but stimulation of STN should further increase GPi activity.

For several years it was therefore assumed that stimulation at the DBS frequencies being used (typically 130 Hz) must have been suppressing activity within the nuclei. This was

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supported by data showing that recording in the vicinity of the cell bodies within STN, when stimulating STN [BGN+00], or recording within GPi with GPi stimulation [BBBG96], demonstrated a *decrease* in apparent cell firing following nearby high-frequency stimulation. Helping to make further sense of this were the findings, in neurons different from the basal ganglia, that synaptic depression could occur at modest stimulation frequencies while the synapses ploddingly worked to repackage neurotransmitter within synaptic vesicles for release [SLBY98]. These findings were all consistent in explaining why DBS had the same effects as ablative lesions. This assumption was unfortunately wrong.

In computational modeling, Cameron McIntyre and colleagues [MGST04] demonstrated that such DBS stimulation close to cell bodies and axons might preferentially initiate action potentials further out along the axons of such neurons than we have appreciated previously, and furthermore that the cell bodies might not reflect these action potentials. Such findings also remind one of the differing requirements in the curvature of a stimulating electrical potential field needed to initiate firing at the terminal end of a neuron process membrane, as opposed to the middle of a membrane such as an axon coursing near an electrode *en passage* [WGD92]. More recently, optical imaging has also confirmed that action potential initiation may be well beyond the axon hillock, in contrast with what we previously assumed [BIA+04].

The definitive experimental evidence that laid to rest the hypothesis that DBS worked by suppressing neuronal activity and creating a *reversible lesion* was the demonstration by Jerrold Vitek and colleagues that such stimulation led to an *increase* in firing frequency in the nucleus receiving the efferent activity from the nucleus being stimulated [HEO+03] as shown in figure 10.6.

So if increasing the frequency of GPi activity through stimulation of STN had the same effect as burning a hole in GPi, the reason had to lie in the dynamics of the effect of increasing the firing rate of GPi. Note two things about the effect of stimulation on GPi activity in figure 10.6: (1) the frequency is increased, and (2) the neural response becomes less episodic and more continuous. The rest of this chapter seeks to understand and exploit this finding.

Increasingly, we run into examples such as this where you cannot possibly understand the raw data recorded from the nervous system without forming a computational model. The model is necessary to create the equations of motion for the network involved. As with Newton's laws, it does not take a very complex system before even the simplest of nonlinear interactions between elements, whether planets through gravity or neurons through synapses, become impossible to put together from visual inspection and guessing.

Jonathan Rubin and Terman [RT04] set out to ask whether the apparent stimulation paradox outlined here was explainable on the *basis of the biophysical properties of the neurons within these networks*. Their goal was "to demonstrate, with a computational model, why this is actually not contradictory, but rather is a natural consequence of the properties of the cells involved" [RT04]. Fitting empirical models to Parkinson's data might produce an

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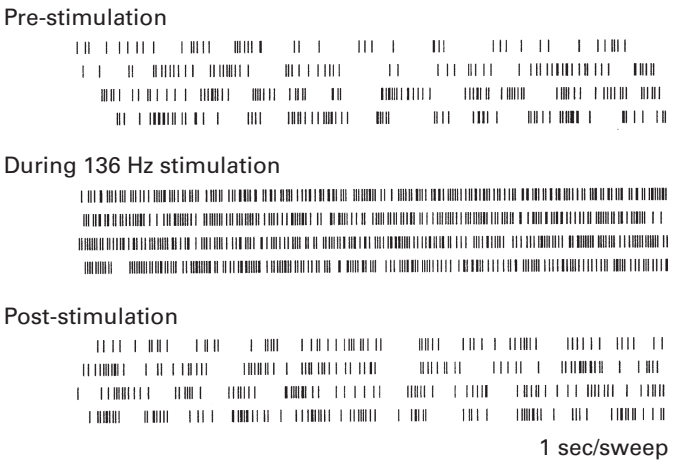


Figure 10.6
Demonstration that 130-Hz stimulation of STN in an MPTP primate increased the GPi cell firing rates. Reproduced from [HEO+03] with permission.

effective controller, but would be no more insightful than proving that you could fit a model to prove that the neurons did what you had observed them to do. The power of model-based control approaches is that we take the insights from what we have learned about these pieces of brain we are working with, and use those fundamental models to guide our observations and control.

10.7 Reductionist Cracking the Deep Brain Stimulation Paradox

We will now extend our model, from the oscillations between GPe and STN [TRYW02], to the effect of these oscillations on the thalamus through the intermediary way station of GPi. The motor structures within the thalamus, with apology to the vast complexity of this organ [GS02], will be viewed as a structure whose task is to faithfully relay information. The thalamus will have two inputs: GPi and sensorimotor signals (figure 10.3).

Rubin and Terman [RT04] distilled the essence of Parkinson's disease symptoms within the output of GPi. In the normal state, these cells fire irregularly and do not interfere with thalamic information relay. In Parkinson's disease, GPi cells fire bursts of action potentials at the frequency of tremor (3 to 8 Hz). The researchers [RT04] assumed that these burst firing cells will exhibit some degree of synchrony in the pathological state. Their hypothesis was that such clustered firing in bursts would impair the sensorimotor relay properties of the thalamic cells.

The sparse structured network (figure 10.4B) from Terman and colleagues [TRYW02] was chosen based on its results compared with the other topologies examined.

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Thalamic cells were modeled with

$$C_m \frac{dv_{Th}}{dt} = -I_L - I_{Na} - I_K - I_T - I_{Gi\to Th} + I_{SM} \tag{10.2}$$

where $I_{Gi\to Th}$ represents synaptic current from GPi to thalamus, and I_{SM} represents sensorimotor input to thalamus. They are of opposite signs because one is inhibitory and the other excitatory.

The specific structures of these currents are of interest to us now, as we will shortly reduce them as a prelude to our control framework.

The leak current is simple

$$I_L = g_L \cdot (v_{Th} - E_L)$$

where g_L is the maximal leak conductance, v_{Th} is the transmembrane voltage on the thalamic cell, and E_L is the reversal potential at which there would be no leak current when $v_{Th} = E_L$. The sodium current is from Hodgkin and Huxley (Hodgkin:1952d):

$$I_{Na} = g_{Na} \cdot m_\infty^3(v_{Th}) \cdot h_{Th} \cdot (v_{Th} - E_{Na})$$

except for the use of Rinzel's approximation [Rin85], substituting m_∞ for m in the sodium gating variable, and in the following potassium current equation, substituting $1 - h_{Th}$ for n as the potassium gating variable (h is the sodium inactivation gating variable)

$$I_K = g_K \cdot [0.75(1 - h_{Th})^4] \cdot (v_{Th} - E_K)$$

The T-type calcium current equation is

$$I_T = g_T \cdot p_\infty^2(v_{Th}) \cdot w_{Th} \cdot (v_{Th} - E_T) \tag{10.3}$$

where $p_\infty(v_{Th})$ is the T-current gating variable, and w_{Th} is the T-current inactivation variable. In these equations, the reversal potentials for leak, sodium, potassium, and T-current are E_L, E_{Na}, E_K, and E_T, respectively. The inactivation variables follow the Hodgkin-Huxley formalism for first-order kinetics:

$$\frac{dh_{Th}}{dt} = (h_\infty(v_{Th}) - h_{Th})/\tau_h(v_{Th})$$

and

$$\frac{dw_{Th}}{dt} = (w_\infty(v_{Th}) - w_{Th})/\tau_w(v_{Th}) \tag{10.4}$$

The sensorimotor current, I_{SM}, was prescribed to be either periodic or, at times, random. The symmetry introduced in this equation by the canceling currents $-I_{Gi\to Th} + I_{SM}$ will

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cause us difficulties, which we will address in section 10.10 (a more complete discussion of symmetries in reconstruction can be found in [SS09]).

These thalamocortical (TC) cells are silent if unstimulated. If stimulated with depolarizing current, they fire progressively faster. However, if they are hyperpolarized, we see progressively more and more intense rebound activity due to the T-current.

We can provide this TC cell with an analog to sensorimotor stimulation by periodically stimulating it. This is a signal that we hope the cell can relay. In figure 10.7A, we start with slow stimulation. The signal is reliably relayed. Now simultaneously we provide the cell with excessive inhibition, such as from an overactive GPi in Parkinson's disease. In figure 10.7B the baseline membrane potential is now more hyperpolarized. With each sensorimotor pulse, the cell rebound spikes because the T-current is deinactivated. This is because hyperpolarization removes I_T inactivation, just as in the sodium inactivation in the Hodgkin-Huxley gating variable h [HH52d]. But removing inactivation is also a relatively slow process. In figure 10.7B, there is sufficient time for this variable to be fully deinactivated. Contrast this with higher stimulation frequencies. In figure 10.7C, the TC cell is reliable, and it remains so in the setting of additional inhibition in figure 10.7D because there is insufficient time for the inactivation to deinactivate.

STN cells were modeled with

$$C_m \frac{dv_{Sn}}{dt} = -I_L - I_{Na} - I_K - I_T - I_{Ca} - I_{AHP} - I_{Ge\to Sn} + I_{DBS} \tag{10.5}$$

Here Rubin and Terman [RT04] added a high-threshold calcium current, I_{Ca}, and the synaptic current is now from GPe to STN, $I_{Ge\to Sn}$. There is also a DBS current, I_{DBS}. The parameters on the STN cells were adjusted so that the cell was spontaneously active, fired at high frequencies when depolarized, and had a less prominent rebound than the TC cell. All of these adjustments were to preserve as much of the qualitative distinction between these cells' firing properties as was observed in prior experiments.

GPe cells were modeled with

$$C_m \frac{dv_{Ge}}{dt} = -I_L - I_{Na} - I_K - I_T - I_{Ca} - I_{AHP} - I_{Sn\to Ge} - I_{Ge\to Ge} + I_{app} \tag{10.6}$$

Input from the striatum was modeled with I_{app}. The GPe cells would continuously fire, initially decrease their firing rate with inhibition, and with more inhibition cluster fire consistent with the results of Terman et al. [TRYW02]. The GPi cells were similarly modeled, except the parameters were adjusted to account for the experimental evidence that these cells tended to fire faster than GPe cells [DeL71].[5]

5. Some further discussion about the applicability of DeLong's results [DeL71] to the classification of human Parkinsonian cellular activity can be found in [SSKW05].

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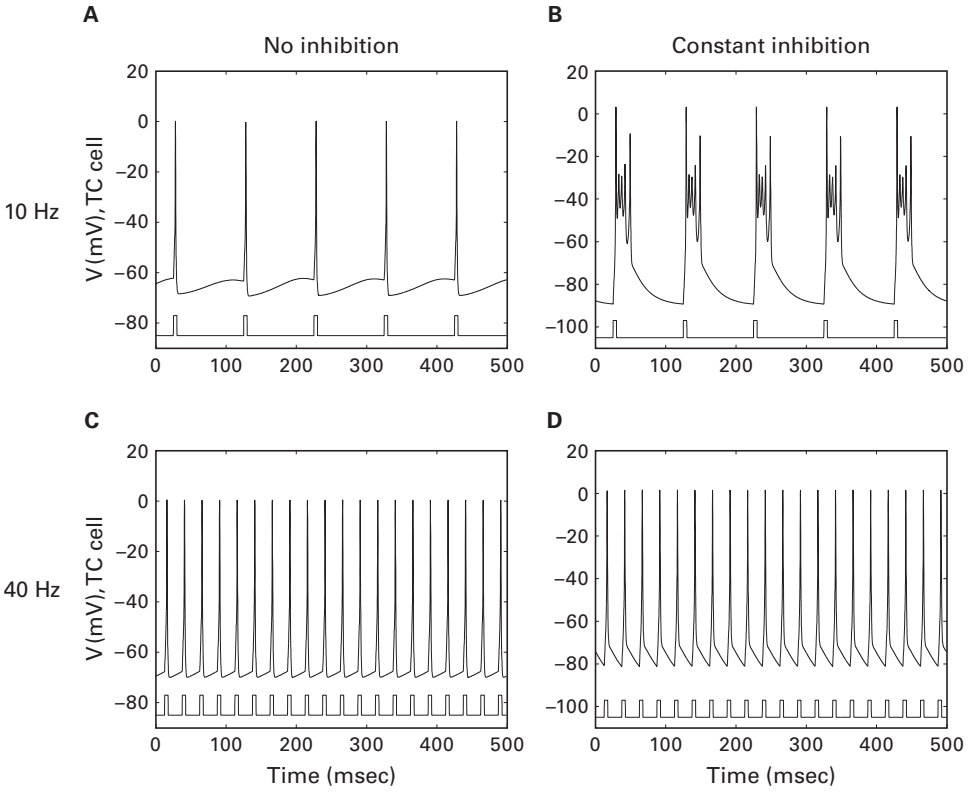


Figure 10.7
Stimulation of TC cell model at 10 Hz (A and B), and 40 Hz (C and D), at low (A and C) and high (B and D) levels of tonic inhibition. Reproduced from [RT04] with permission.

Following the schematic in figure 10.5, the Parkinsonian state is recreated by increasing the striatal input to GPe, and decreasing the amount of internal recurrent inhibition within GPe. The result is that the normal reliability of the TC cell to transmit sensorimotor information, illustrated in figure 10.8, becomes impaired in the Parkinsonian state. The key quantity here is the error rate of transmitting sensorimotor input into TC spikes. An error index can be created as

$$\text{Error Index} = \frac{\text{Missed spikes} + \text{Bad spikes}}{\text{Total inputs}}$$

$$= \frac{\text{False negative} + \text{False positive}}{\text{Total inputs}}$$

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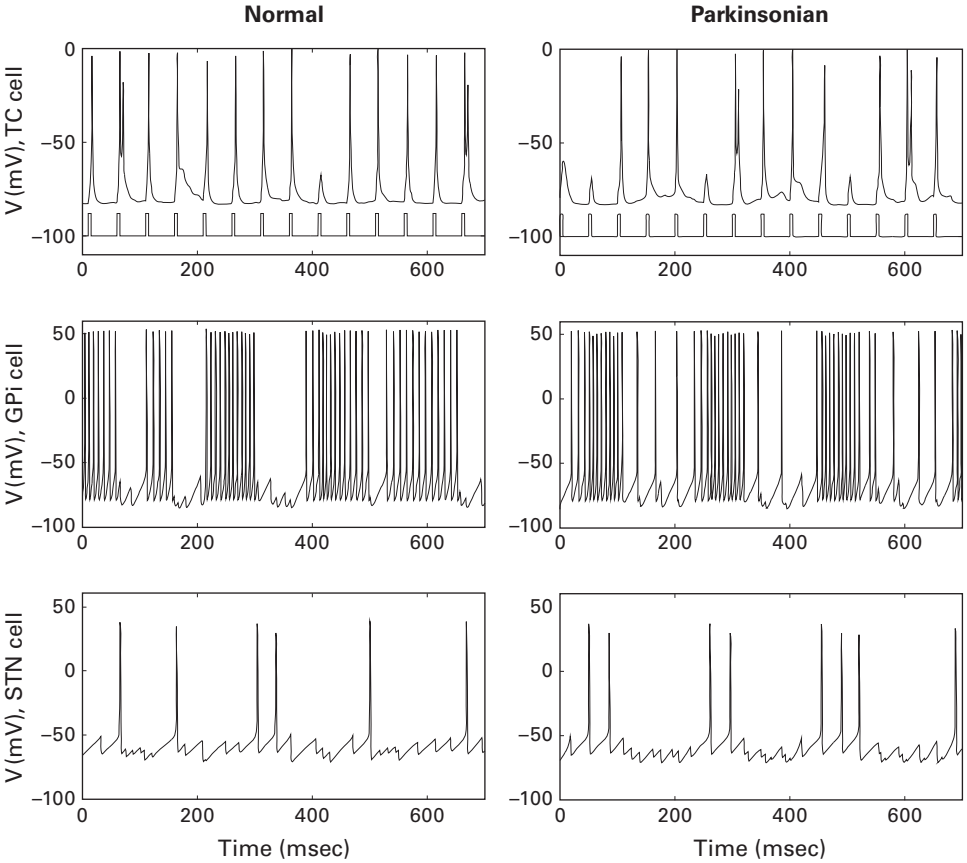


Figure 10.8
Model of the TC (upper panels), GPi (middle panels), and STN (lower panels) cells to periodic sensorimotor stimulation (top panel) in the normal (left) and Parkinsonian (right) states. Reproduced from [RT04] with permission.

Rubin and Terman [RT04] showed that the error rate was significantly elevated in the Parkinsonian[6] state compared with the normal state, and that the error rate could be normalized by simulating DBS using a constant level of high-frequency stimulation of the STN. The key to understanding these results is to know what the TC cell is receiving. In the Parkinsonian state, the amount of inhibition is fluctuating more than normal. This induces sequential excess suppression and rebound bursting in the TC cell, which destroys reliability. Applying DBS decreases the fluctuations that the TC cell receives, despite an overall increase in GPi cell firing.

6. It is semantically sensible to designate dynamics similar to those seen in Parkinson's disease as "Parkinsonian," and to similarly term analogous models as Parkinsonian.

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A key insight of Rubin and Terman [RT04] was that the qualitative features of the above could be preserved in the TC cell *without* fast dynamics.[7] Taking their TC membrane model, they removed the fast Hodgkin-Huxley currents for sodium, I_{Na}, and potassium, I_K, and created a two-variable description of TC cell membrane dynamics:

$$\begin{aligned} \frac{dv}{dt} &= -(I_L + I_T)/C_{Th} - I_{Gi\to Th} + I_{SM} \\ \frac{dw}{dt} &= \varphi \cdot (w_\infty(v) - w)/\tau_w(v) \end{aligned} \tag{10.7}$$

In equations (10.7), v represents membrane voltage, φ serves as a relative rate constant between the two differential equations, w represents T-current inactivation, and the availability of T-current I_T will be the key to reliability.

The current $I_{Gi\to Th}$ will be modeled as a true synaptic inhibitory input onto the TC cell as

$$I_{Gi\to Th} = g_{Gi\to Th} \cdot s_{Gi} \cdot (v - E_{Gi\to Th}) \tag{10.8}$$

where s_{Gi} will be a small positive constant in the normal, a periodic square wave in Parkinsonian, and a larger positive constant under conditions of DBS. But because a reversal potential is attached to all synaptic ionic channels, $E_{Gi\to Th}$, the current $I_{Gi\to Th}$ will fluctuate with voltage v even when s_{Gi} is a constant. The equations for the T-current, I_T, are as given in equations (10.3) and (10.4).[8]

The beauty of the two-variable reduction of the TC cell is that the nullclines can be plotted in two dimensions and visualized,[9] providing further insight into the dynamics. It is also a simpler system to fit, if you are interested in observing actual data or developing control laws, than using the full model.

The nullclines for v and w are shown in figure 10.9A. In excitatory cells and their models, there is almost always a cubic or N-shaped nullcline for the fast excitatory variable, the voltage v in this case (see chapter 4). Keep in mind that in this reduced model there are no true action potential spikes (no I_{Na} or I_K currents). These phase space plots show us the slow dance between voltage changes, v, and I_T inactivation, w. In the middle panel, we see the effect of DBS. Increasing the synaptic current from GPi, s_{Gi}, as the DBS parameter

7. This is only one example of a case where slow excitability dynamics may be more important than the faster spiking dynamics in determining pathological activities in dynamical disease. In recent work, we have shown that in models of epileptic seizures, relatively slow fluctuations in potassium dynamics may underlie the more complex spike dynamics seen in individual cells [CUZ+09, UCJBS09].

8. The square on p_∞ can be dropped in the reduced model. See [RT04] appendix for all equation details.

9. Nullclines are the curves where the rate of change of the variables are equal to zero. The intersection of more than one nullcline gives us the solutions, stable or unstable, of the system variables.

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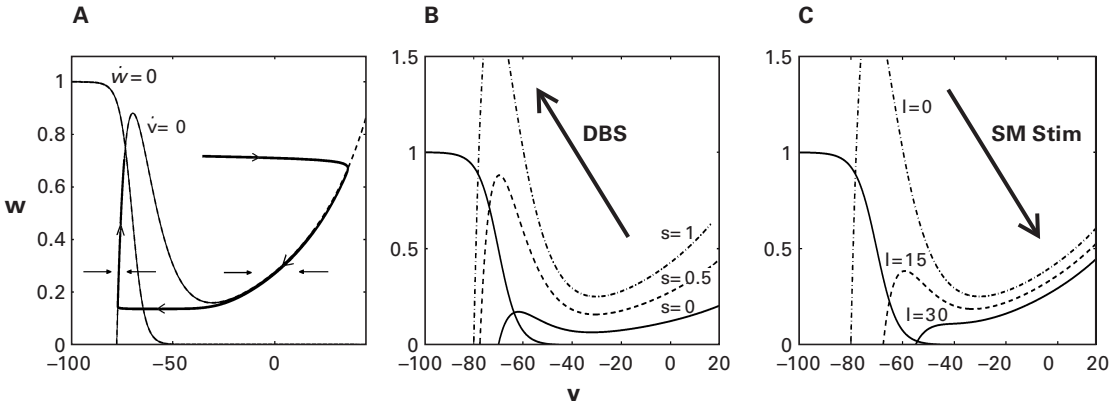


Figure 10.9
Nullclines for $\dot{v} = 0$ and $\dot{w} = 0$. In A, the heavy line is a trajectory initiated by depolarizing the rest state at the intersection of the nullclines from −65 mV to −25 mV, and the subsequent trajectory in the phase plane closely tracks the faster equilibrating v nullcline, and slowly works its way up along the direction of increasing w as the T-current deinactivates (that is, the availability of T-current, w, *increases*). B shows the effect of DBS, elevating the v nullcline. C shows the effect of sensorimotor stimulation, SM, which lowers the v nullcline. Reproduced from [RT04] with permission.

in (10.8) literally adds the $I_{Gi\to Th}$ current term in (10.7) to the solution of the v nullcline.[10] This has a qualitatively opposite effect to increasing excitatory sensorimotor stimulation I_{SM} in (10.7), which decreases the height of the v nullcline. At the point where these nullclines intersect is the resting steady state for v and for w. The T-current inactivation, w, is a key factor in whether this system will respond with a rebound burst, respond reliably to a sensorimotor input, or be inactive and unreliable by not responding at all.

When the intersection is brought lower on this phase plot, the system requires enough T-current, by having high enough w, to be able to push over the peak in the $\dot{v}$ nullcline, the *knee* of this curve (the upgoing bump). As sensorimotor stimulus steadily increases, the value of w gradually decreases setting the system up for trouble.

What this means is that if the position of the intersection of the nullclines is such that a stimulus (I_{SM}) comes when the value of w is above the knee of the $\dot{v}$ nullcline, then the cell can fire a spike. If the system were subjected to *episodic* bursts of inhibition through a Parkinsonian GPi input to TC, then when the intense inhibition is released the cell would rebound. But if you hit the cell with an intense enough *steady* DBS input, the cell will

10. The reason is that when $dv/dt = 0$, the solution of v is

$$v = (I_L - I_{Gi\to Th} + I_{SM})/(g_{Ca}\, p_\infty (v - E_{Ca}))$$

where the very high positive reversal potential for calcium, E_{Ca}, always makes $(v - E_{Ca})$ negative. Because of the negative reversal potential $E_{Gi\to Th}$ for the inhibitory synapses, $(v - E_{Gi\to Th})$ is almost always positive in equation (10.7). So the net effect is that increasing $I_{Gi\to Th}$ leads to an *increase* in the v nullcline.

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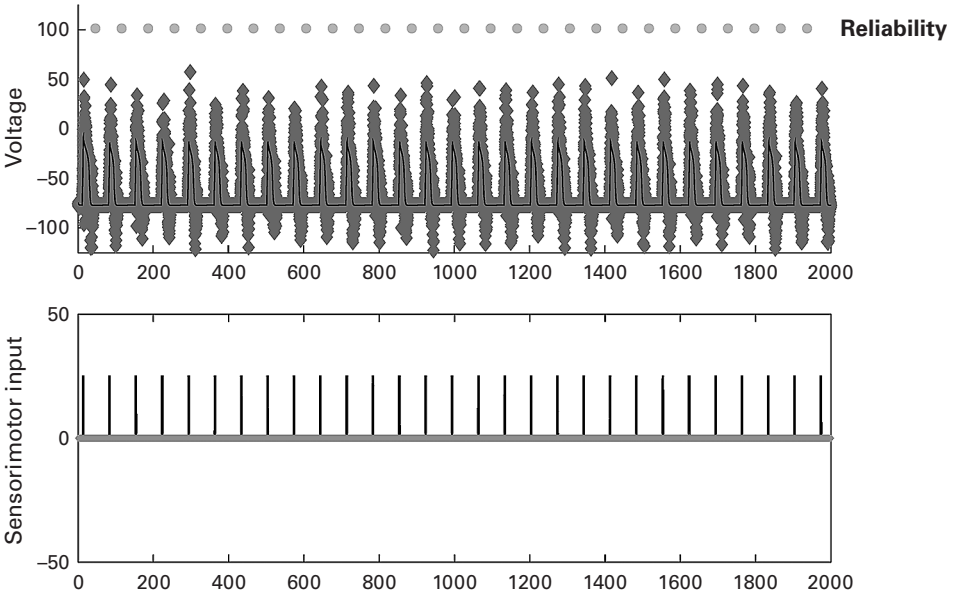


Figure 10.10 (plate 23)
Normal response of TC cell to periodic sensorimotor stimulation. Voltage in reduced TC cell model (thin black line, upper panel), with substantial added noise to serve as a noisy observable (blue markers, upper panel), and sensorimotor input (red pulses, lower panel). Each time a sensorimotor pulse is reliably transmitted, a marker (green) is placed above the successfully transmitted spike. This cell is 100% reliable. Reproduced from [Sch10] with permission.

maintain sufficient T-current and has no opportunity to rebound (there is no abrupt turn-off of inhibition). Because in the Parkinsonian state the T-current availability fluctuates on and off, the result is tremor at this fluctuation frequency. DBS can level this out and simultaneously keep the cell in a state ready to fire reliably to excitatory inputs by keeping w moderately high.

Now let's look deeper into these reduced dynamics in normal, DBS, and Parkinsonian states. In figure 10.10 (plate 23) we see a normal state. The input to the reduced TC cell from the GPi is a modest constant, which represents a state where there is asynchrony among the GPi cells, and there results a modest steady level of inhibition to the TC cells. Periodic sensorimotor stimulation is represented by brief square wave excitatory inputs (second panel). A substantial amount of random measurement noise is added to the actual TC voltage. These noisy measurements are what we record from in these models as our *observable* variable. Note that for each sensorimotor stimulus, a "spike" is transmitted from the TC cell. This is reliable transmission (100% reliable in this case).

The DBS state is modeled as increased tonic inhibition on the TC cell. The Parkinsonian case can be idealized as a periodic fluctuation in the GPi input onto the TC cells. In [RT04],

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such an effect can be shown as a slow on and off square wave modulation of GPi output. Using the reduced model, when excess inhibition turns on, w gradually increases, T-current becomes more available, and spiking becomes more reliable after an initial suppression by the inhibition. When inhibition suddenly stops, w remains too high, and an excessive response, with a period of rebound excitation, occurs. During the trough of the Parkinsonian inhibition fluctuations, spikes are reliably transmitted but w gradually inactivates, and when the next inhibitory pulse hits, there is spike failure. This sequence of recurrent spike failure and rebound creates unreliability. Recall that in the reduced model there are no fast sodium spikes—the slow "spikes" are periods of increased *excitability*; if we had I_{Na} and I_K, we would get a burst of fast sodium spikes riding on the excitability rebound event, and this would also reflect unreliable information transmission. Steady DBS, literally taking out the troughs in the Parkinsonian fluctuations, can restore reliability, as we will shortly examine in more detail.

10.8 A Cost Function for Deep Brain Stimulation

Xiao-Jiang Feng and colleagues sought to further explore the work of Rubin and Terman by introducing *optimization* principles [FSBG+07, FGR+07]. They, like Rubin and Terman [RT04], employed the sparse structured network of Terman and colleagues [TRYW02].

Why not just use the high-frequency stimulation that presently demonstrates efficacy in clinical use? For one thing, the more energy we apply per day, the sooner the batteries of the stimulators wear out and the devices need to be surgically replaced.

But beyond batteries, as stimulation intensities increase, so will negative cognitive side effects [AVRH+08].[11] I would pose a general principle that in the treatment of any dynamical disease of the brain, one needs to optimize the beneficial effects of symptom relief (e.g., tremor), with the inherent effects of increasing cognitive dysfunction with increasing DBS energy.

In addition, it is not trivial at all to make clinical adjustments to a patient's stimulation parameters. Once set, the parameter space of stimulation intensity and frequency, along with duty cycle, is enormous. Making changes, waiting several days or weeks to see the steady-state effect, and trying to optimize settings is difficult in patients. So if a stimulator is working and showing benefit, optimizing in this ad hoc fashion is generally avoided. Nevertheless, there is evidence that some patients' symptoms can be improved at frequencies different from the standard starting frequency of 130 Hz [MDD+08]. And who is to say that a more

11. There are recent reports that steady low-amplitude DC stimulation to some parts of the brain might improve cognitive capabilities [MHMB06]. But this is a polarization effect on certain parts of the cortex. To the author's knowledge, all pulse stimulation of deep brain structures will have negative effects on cognitive information processing as stimulation is increased. This will likely apply to DBS treatment of cognitive disorders, such as depression and obsessive compulsive disorder, as well as to Parkinson's disease.

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complex nonperiodic stimulus delivered in open-loop might not be more beneficial than the standard periodic ones?

Last, the beneficial effects of DBS over the long term in a patient with a neurodegenerative disease (and perhaps those without degeneration) is a moving target. Patients with DBS treatment of Parkinson's disease continue to deteriorate within the tempo of progression of Parkinson's disease [KBVB+03]. Neural networks learn and change in response to stimulation. A means to automatically adapt and optimize such stimulation over time will be a very valuable advance in our stimulation technology.

One requires metrics for optimization. Feng and colleagues introduced a *reliability* measure similar to Rubin and Terman [RT04], as well as a GPi cell *correlation* metric. Neither metric has a clear route to practical application in human subjects using our present optimization done *by hand*. But these tools can be immediately applied if we construct a customized computational model for an average or specific patient. And this is the model-based approach that can be employed in future automated closed-loop optimization.

The structure of their cost function was [FSBG+07]

$$J^i = x^i + wR$$

x^i = criterion i (reliability or correlation)

R = energy required

where w is a weighting parameter. They employed a genetic algorithm to search their parameter space and accomplish optimization.

One way of implementing such a cost function is to integrate the current coming out of the stimulator, I_{DBS}, and subtract this weighted integral from the reliability, Rel, as

$$J = Rel - w \int_{t=0}^{T} I_{DBS} dt$$

So starting from a sparse structured network, with two TC cells, Feng and colleagues replicate the fundamental findings of Rubin and Terman [RT04] in figure 10.11A–F.

They then use their genetic algorithm to explore the parameter space of frequency and amplitude as shown in figure 10.11G. In the far rear corner of figure 10.11G lies the high-frequency and high-amplitude parameter regime typically employed in clinical use, and the reliability there is high. But notice the other peaks. The most prominent secondary peak is between 40 and 50 Hz, and indeed, there is some clinical evidence that this frequency range might be therapeutic [MDD+08]. Figure 10.11H shows the results of a minor peak at a period of 80 ms (12.5 Hz); there is actually some clinical study of stimulation in the range of 5 to 20 Hz, which was (unfortunately) singularly unimpressive [ECL+08]. We note that another exploration of frequency in an extended model of Rubin and Terman [RT04] demonstrated that the frequencies greater than 100 Hz were most effective [PRSC08].

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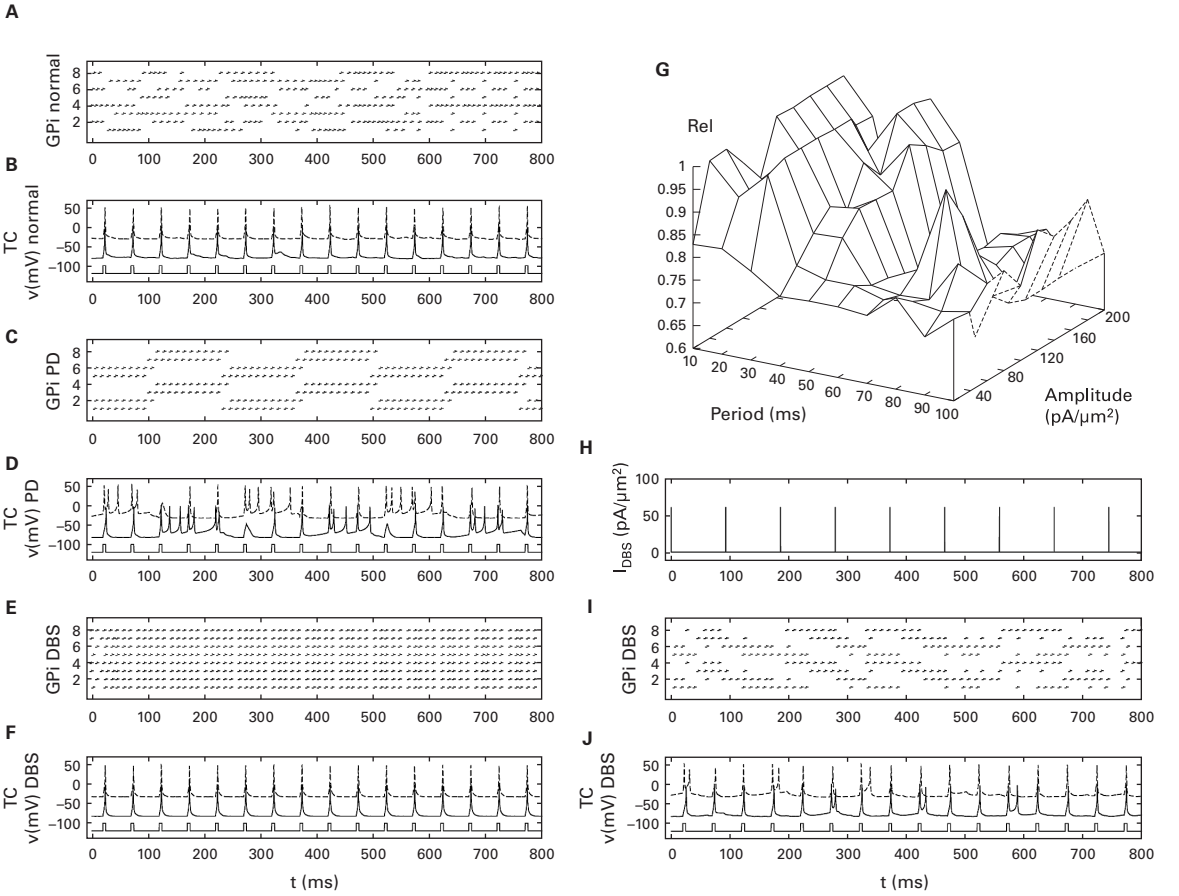


Figure 10.11
Exploration of the parameter space of period and amplitude focusing on reliability: Normal (A,B), Parkinson's disease (PD) (C,D), and DBS in PD (E,F). Optimization of reliability as a function of period and amplitude (G). Slower DBS frequency at 80 ms reliability (Rel) peak in H. Reproduced from [FSBG^{+}07] with permission.

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Reliability is an interesting measure. There seems no way to directly measure in real life the quantities needed to calculate reliability: sensorimotor inputs and thalamic relay cell output in response to this input. But such inputs can be readily modeled using a data assimilation framework. Feng and colleagues [FSBG+07] also employ a correlation measure, which was a combination of autocorrelation and crosscorrelation among GPi cells. This would require a microelectrode array to be placed within GPi. Such microarray technology is now becoming available on the shafts of clinical DBS electrodes. A simpler way to infer such correlation in practice would be to estimate spectral concentration in GPi local field potentials, and this is directly observable from an ordinary DBS electrode placed within GPi.

An important contribution of Feng and colleagues [FSBG+07] is that they demonstrated in this computational model that stochastic stimulation as well as complex waveforms can demonstrate potential efficacy.

10.9 Fusing Experimental GPi Recordings with DBS Models

A creative next step was performed by Guo and colleagues [GRM+08]. They took recordings of GPi spike timings from normal and MPTP primates, with and without DBS. They applied a structure function to these spike timings to recreate an estimate of the synaptic currents produced in the thalamic relay cells from such spiking. They used the model from Rubin and Terman [RT04] to estimate the reliability of the transmission of spikes through the thalamus. Their results demonstrate that the error index (described earlier) is higher for the MPTP monkeys without DBS or with subtherapeutic DBS, and consistently lowest for normal monkeys and those MPTP monkeys with therapeutic levels of DBS.

This is another example of how we can apply models in real life applications to Parkinson's disease treatment. Recording of an array of GPi cells would be required. One could then follow the prescription of Guo and colleagues [GRM+08] to estimate ongoing reliability, and potentially perform closed-loop optimization.

A very interesting insight from Guo and colleagues [GRM+08] was to *reverse correlate* what the inhibitory current drive from GPi was *prior* to a spike. Spikes can be faithful in following a sensorimotor input, they can miss their chance to follow, or multiple bad spikes can ensue. Guo and colleagues found that missed spikes correspond to a relatively rapid rise in inhibition, and bad spikes correspond to a relatively rapid decrease in inhibition. This is exactly what one might have expected from the reduced model effects illustrated in figure 10.9.

10.10 Toward a Control Framework for Parkinson's Disease

Parkinson's disease will be the first *dynamical disease* [MG77] amenable to management through model-based feedback control systems.

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First, examine the schematic in figure 10.3. We can now relatively safely place electrodes in the STN, GPi, or thalamus.[12]

What about the small amount of damage that we get from just passing a 1 to 2 mm-thick depth electrode? Recall that in Parkinson's disease, making a lesion in these small structures is clinically beneficial. Amazingly, we observe that in the course of placing DBS electrodes in Parkinson's disease patients, there is an immediate clinical improvement in about *half* of patients at the conclusion of electrode placement, *before* the electrode is connected to the stimulator [Tas98]. This has been termed the *microthalamotomy effect* [Tas98]. A nearly identical experience has been reported in the placement of DBS electrodes in small thalamic targets for epileptic seizure suppression [HWDL02]. *Our DBS treatments are likely a combination of small lesions overlain with chronic electrical stimulations*.

Despite the beneficial effect of some of the microlesions created by electrode insertion, the best clinical strategy is almost certainly to keep the number of electrodes inserted to the absolute minimum.[13] But the network that is important in Parkinson's disease control is spread out over at least four separate structures: STN, GPe, GPi, and thalamus.

From a clinical perspective, the GPi would likely be an easier and safer target than the smaller STN or thalamic targets. A small hemorrhage or lesion effect in GPi has a reasonable chance of actually benefitting some of the Parkinson's disease symptoms. But with a superb animal model in the MPTP primates, exploring which nuclei are optimal to record data from, and work out the control algorithms, offers a valuable way to learn how to build such systems prior to their implantation in human patients. Nevertheless, given that our clinical opportunities with routine implantation in patients is ongoing, the chance to take advantage of existing STN or GPi human placements in algorithm development is highly attractive.

The models we have just discussed permit us to sample from a single nucleus within this network and reconstruct what the remainder of the network is doing. Our present state-of-the-art models appear sophisticated enough to consider using these models in a data assimilation framework. And the technology to perform chronic real-time sampling from these structures is presently available and in clinical use.

So let's begin to outline a control theoretical framework from what we have discussed.

12. The best current data, from A. Benabid (2007, personal communication), suggests an 8% risk of a complication per depth electrode pass. Half of these events (such as small hemorrhages) are asymptomatic. Half of the remaining give transient effects. A reasonable estimate is that 2% of electrode passes into the brain using modern stereotactic techniques will result in a significant neurological complication [LGT09].

13. This is a conjecture by the author, and one can argue in the abstract that we are accumulating a tremendous body of data showing that microlesions plus electrical stimulation through single electrodes is efficacious. But we have no such body of data suggesting that multiple microlesions in separate unilateral structures will be beneficial. And from a surgical standpoint, every electrode insertion increases operative time, increases the risk of hemorrhage and neurological deficit, and increases the risk of infection. Surgical minimalism should be a guiding principle in the absence of strong evidence supporting the contrary.

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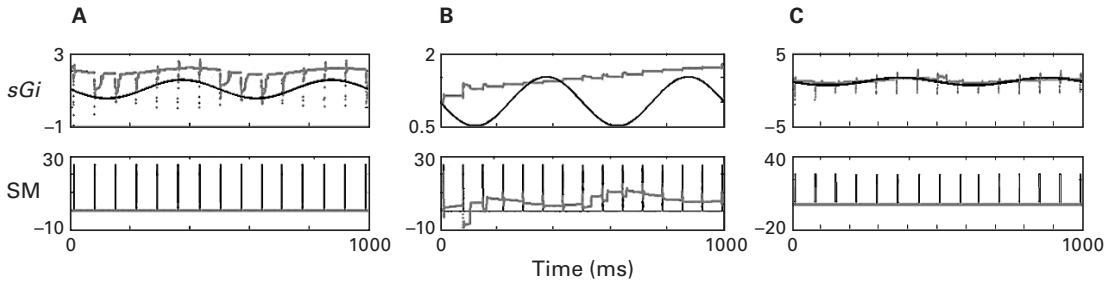


Figure 10.12 (plate 24)
Tracking and estimating parameters as a function of process noises Q. Estimates of synaptic current from GPi (sGi, upper), and sensorimotor input (SM input, lower). Process noise parameters are (A) $sGi\ Q = 30$, $SM\ Q = 0.01$, (B) $sGi\ Q = 0.01$, $SM\ Q = 30$, and (C) $sGi\ Q = 10.0$, $SM\ Q = 0.01$. The algorithmic incorporation of such process noise can be explored in the code archive with [SS08]. Reproduced from [Sch10] with permission.

The first issue with parameter estimation for equation (10.7) is to notice that there is a symmetry with the terms $(-I_{Gi\to Th} + I_{SM})$. The tracking algorithm will simply apportion such current equally between these two sources. One option is to combine the currents into one sum, which physically is what is happening to the TC cells. But this defeats our purpose here, where we would like to estimate quantities such as $(-I_{Gi\to Th})$ in isolation. A more extensive discussion of symmetries in such equations can be found in Sauer and Schiff [SS09] (see chapter 4).

When faced with such symmetries, it is best to get rid of them. If that is not feasible, then an empirical rule of thumb seems to be to set process noise in rough proportions to the average magnitudes of the corresponding variables. One could adaptively tune these process noises over time by tracking innovation error. Process noise, Q, is uncertainty commonly added to the model of the process (the *plant* in control jargon) in analogous applications (chapter 2). In this particular instantiation of ensemble Kalman filtering, we will use several Q values as the assumed variance in the respective parameters to be tracked. In addition to apportioning variance to the respective parameters, this also has the benefit of preventing a Kalman filter from driving the parameter covariance to zero.[14] In figure 10.12A, the Q for $(-I_{Gi\to Th})$ is 30, while the Q for I_{SM} is 0.01. Note that the rhythmicity of $(-I_{Gi\to Th})$ is resolved, while none of the features of I_{SM} are picked up (figure 10.12A). Now reverse the situation, setting Q for $(-I_{Gi\to Th})$ to 0.01 and the Q for I_{SM} to 30; the $(-I_{Gi\to Th})$ will be poorly tracked, but the sensorimotor inputs I_{SM} are better tracked (figure 10.12B). A more balanced set of process noises, where Q for $(-I_{Gi\to Th})$ is 10, and Q for I_{SM} is 0.01, yields a more optimal tracking of $(-I_{Gi\to Th})$ (figure 10.12C) (see plate 24.).

14. A detailed discussion of process noise can be found in [Sim06a].

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We can calculate a running *reliability* index over the past 10 msec (ignoring for now having to objectively define "bad" spikes unrelated to sensorimotor inputs) as

$$\text{Reliability} = \frac{\text{Number of spikes that got through (within 10 msec)}}{\text{Total number of impulses}}$$

Let's now focus on the heart of the dynamics critical for the Parkinsonian state—the T-current inactivation w. This is a rather independent variable, in time scale and with respect to symmetry, and lies at the heart of the issues of unreliability.

In figure 10.13A (plate 25), I employ the reduced TC cell model from Rubin and Terman [RT04] in a data assimilation framework. The observable will be noisy voltage from the TC cell in the upper panel of the figure. Below that is shown the reconstructed estimated T-current inactivation, w, and the reconstruction of estimated GPi and sensorimotor activity (panels 2 through 4, respectively). The techniques used to accomplish this reconstruction are detailed in [SS08, SS09, US09], and a code archive for the basics of such data assimilation can be found in supplementary data with [SS08].

One could similarly record the average GPi output from a DBS electrode and reconstruct the estimated TC cell activity.

The sensorimotor input, in the fourth panel of figure 10.13, causes the model TC cell to generate or fail to generate reliable spike activity in the upper panel. I build up a reliability index by averaging the last ten sensorimotor spike responses, and plot this at the very top of the upper panel.

It is tempting to examine and consider control from the nullclines. There is an extensive literature in control theory on what is generally termed *variable structure control*. First described by Utkin in the 1970s [Utk77], this control law strategy is now more commonly referred to as *sliding mode* control [DZM88, YUO99]. We could use such nullclines to generate control functions toward which we seek to target the system. In figure 10.14A, we see the actual nullcline intersection of our simulation of the reduced TC cell as the GPi current switches from off to on (refer to figure 10.9). The w nullcline does not change here as GPi input fluctuates. In figure 10.14B, I show an estimation of these nullclines from a reconstruction of these curves using an unscented Kalman filter. There is a great deal of uncertainty in the v nullcline. So estimating a control surface in this phase space is not a trivial problem that we know how to solve at present.

But recall that, as discussed Rubin and Terman [RT04], it is not the nullcline intersection so much as the value of w that is the most valuable single feature in explaining the dynamical response of the TC cell. So let's estimate w by itself.

In the second panel of figure 10.13A is the estimate of the reconstructed T-current availability w (red line). Note that w increases when the GPi current increases in the third panel (solid black line). I have deliberately adjusted the ratios of the process noises so that the sensorimotor input is *not* tracked well (fourth panel), in order to minimize the symmetry

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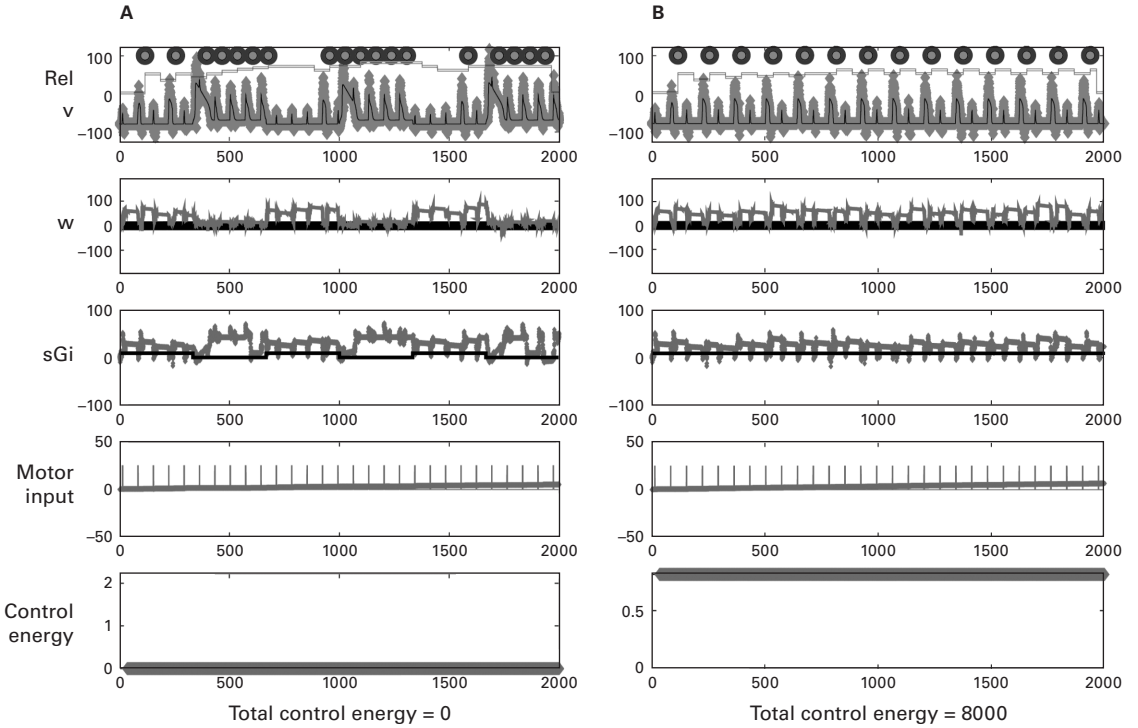


Figure 10.13 (plate 25)
(A) Uncontrolled reduced TC cell dynamics in the Parkinsonian state with fluctuating current from GPi (sGi). (B) Perfect DBS stimulation filling in the troughs in the fluctuating current from GPi. Top panels show noisy observable voltage (blue symbols), reliability as piecewise continuous plots without (blue) and with (red) control on, the green circles are the timing of SM spikes, and the smaller red circles are transmitted spikes. The second panels show estimated w (red). The third panels show real (black) and estimated (magenta) synaptic current from GPi (sGi, estimated values multiplied by 10 for discriminability from the true values). The fourth panel shows real (red) and estimated (magenta) motor input (we are deliberately not trying to reconstruct motor input in the reconstruction through Q ratio adjustment). The bottom panel shows the running control energy, the squared value of the control signal at each time point, and the total sum of squares given as total control energy. Reproduced from [Sch10] with permission.

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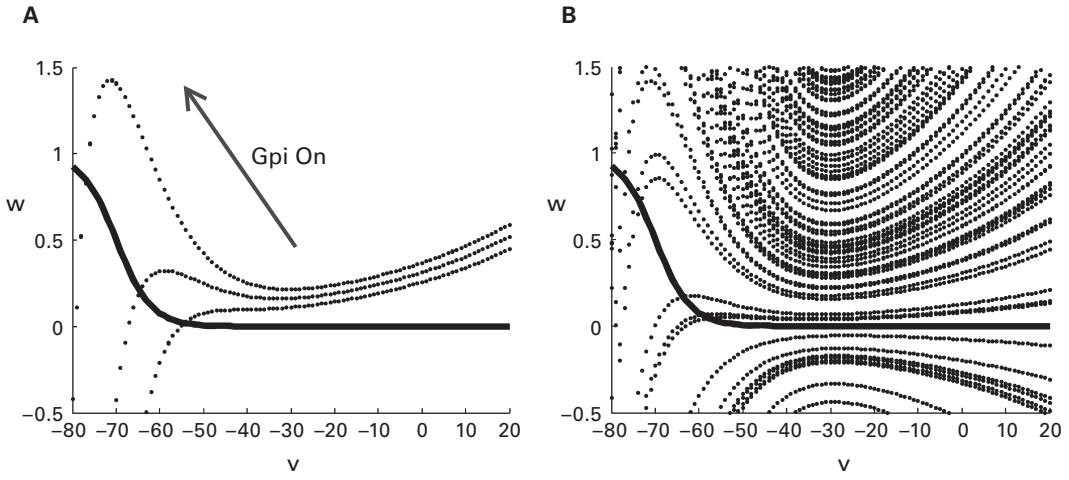


Figure 10.14
Nulclines for w (thick line) and v (dotted line). In A, with increased GPi inhibition onto the reduced TC cell, the v nullcline is elevated with respect to the w nullcline. B shows reconstruction estimates of the nullclines from noisy measurements. Reproduced from [Sch10] with permission.

of this current in the voltage equation. The bottom panel shows the running plot of control energy, which is zero for this example without feedback.

Now let's use *perfect* deep brain stimulation. We will, in the sense of Rubin and Terman [RT04], fill in the gaps of the fluctuating synaptic current from GPi, and show these results in figure 10.13B. Providing a steady level of DBS increases the amount of inhibition arriving on the TC cell. This has the effect of removing the fluctuations in inhibition creating the gaps in reliability shown in figure 10.13A, but it also dampens down the responsiveness of the TC cell. Only about half of the incident sensorimotor spikes get through reliably. And the cost, in terms of control energy, is high (8,000 on an arbitrary scale). On the other hand, the fluctuations in w are reduced, and we maintain an overall high level of w (second panel figure 10.13B).

Now let's use adaptive feedback based on the estimated w. In figure 10.15A (plate 26), I show the effect of an optimal amount of proportional feedback gain based on a moving average (35 msec) of the estimate of w. This feedback is very effective in restoring most of the unreliable (missing) spikes. The total cost in control energy is about half of the perfect DBS case shown in figure 10.13B. Now let's present a more realistic DBS scenario than in figure 10.13B, one in which a constant stimulation (open loop) will be *added* to the fluctuating Parkinsonian GPi signal. Figure 10.15B shows the largest (and most effective) additive current that is stable in this model. Here we are limited by the relatively large peak fluctuating GPi currents being applied already in the Parkinsonian state, as the dynamics of the TC cell become unstable if the impinging currents become excessive. The figure shows

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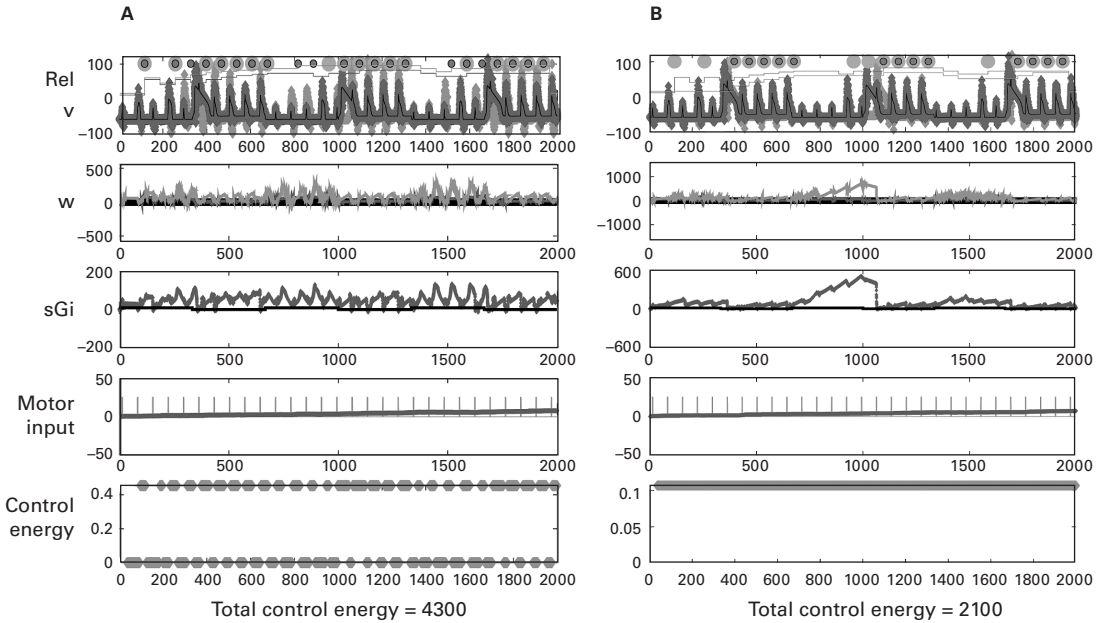


Figure 10.15 (plate 26)
Feedback control scenario based on turning on and off deep brain stimulator based on a running average of the estimated T-current availability shown in A. B shows a scenario where a constant amount of deep brain stimulation is simply added to the fluctuating Parkinsonian GPi output. No adjustment of GPi constant stimulation comes close to the reliability achieved with the closed loop feedback scenario shown in A. Symbols as in figure 10.13. Reproduced from [Sch10] with permission.

that such constant DBS does not appear capable of achieving the reliability possible with feedback control.[15]

It is important to note that in this simple scenario, a range of additional adjustable parameters are important. First, there is the ever-present issue of covariance inflation [AA99].[16] In figure 10.16A, we see that adjusting the small covariance inflation parameter has a substantial effect on the reliability of the adaptive system. Similarly, the gain on the feedback control is important. In Figure 10.16B, we see that optimizing gain readily reveals a region

15. In real life, such model instability would not arise. So long as we stay within the safety limit of the electrical stimulation, we might be able to deliver more GPi current to the TC cell. But the reliability steadily decreases in this scenario as this type of additive DBS increases in the model, since too much inhibition shuts off the TC cells. We find no inference that such open loop stimulation is going to be as effective as an intelligent adaptive approach as illustrated in figure 10.15A.

16. Recall that even Julier and Uhlmann [JU97a] had an adjustable inflation parameter that they required to tune their original unscented filter. As in so many of these recursive filter designs, there is a trade-off between tracking accuracy versus stability with greater uncertainty. One also needs to counter the tendency of these filters to recursively drive the uncertainty to zero, which ruins tracking ability.

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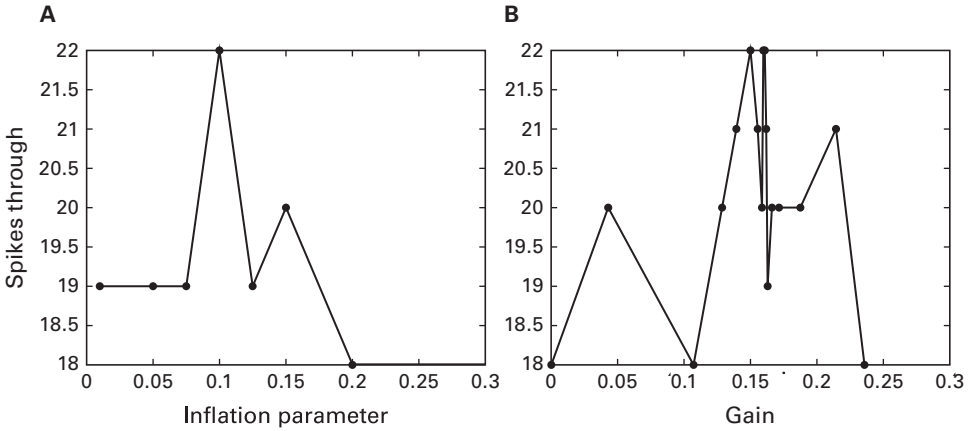


Figure 10.16
Optimizing reliability as spikes that are transmitted reliably through the TC cell as a function of the covariance inflation parameter (A) and proportional control gain parameter (B). Reproduced from [Sch10] with permission.

where spike throughput is best. Both of these functions are not smooth, and implementing such an algorithm should be done with continual adaptation of such parameters based upon the monitoring of the system performance.

Another alternative is to generate a control signal based on the estimated GPi output, shown in figure 10.17 (plate 27). Since we know the control signal added, we can subtract this to follow just the underlying estimated GPi input to the thalamus. Since the estimates of GPi input to thalamus are noisy (dotted blue line in figure 10.17), it is helpful to create a moving average filter of this estimate (solid blue line) to prevent the controller from turning on and off too often. Since at the heart of Parkinson's disease physiology are the large-scale slower fluctuations, we can create a long-term running average of the GPi output, much longer than the noise-reducing short-term moving average, and let this serve as an adapting threshold (magenta line). Control is turned on whenever the short-term moving average (blue solid line) falls below the long-term moving average (magenta line). The control is applied in this case by turning on the stimulator with the same constant amplitude, shown as the red lines. The underlying true GPi output fluctuations are shown as a black line for comparison. Note that a *control reliability* can be calculated as the fraction of time that the control signal (red) is on or off in correct reflection of the peaks and valleys in the true GPi signal (black). In this example, the control reliability is 67%. But the effect on the neuronal reliability in the TC cell, our goal, is not very impressive (a few additional transmitted spikes in the controlled case, but also some missed spikes).

Since reliability is our goal, and since we are estimating it, why not use it as the control parameter? Recognize that the way I have been calculating TC cell reliability in figures 10.14 through 10.17 employed a moving average of relatively infrequent events.

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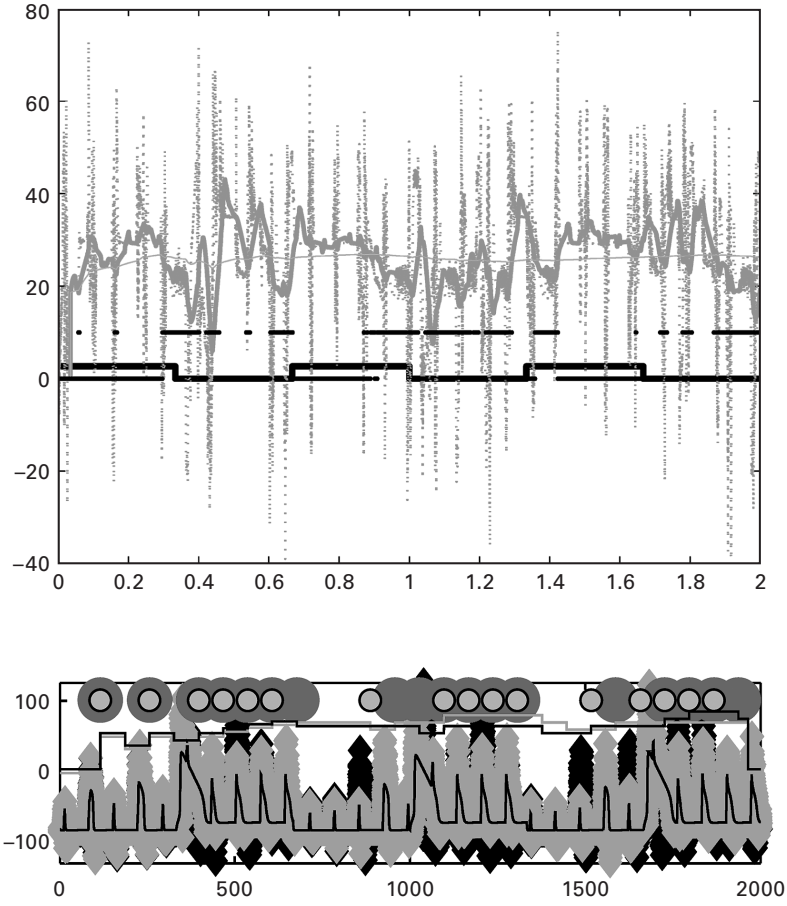


Figure 10.17 (plate 27)
Estimated GPi input to TC cell (blue dotted line), and smoothed short-term moving average of this GPi input (blue solid line). We take a long-term moving average of this current (magenta line) as an adapting threshold to tell when the more instantaneous GPi input is fluctuating up or down. Crossing below the threshold determines when to turn the control on (red). The actual GPi fluctuations are shown (black lines). In the lower panel are the results with control off (blue markers), and on (red markers), and the uncontrolled (green markers) and controlled (magenta markers) spikes transmitted. The running reliability of the TC cell is plotted as a piecewise continuous line for uncontrolled (blue line) and controlled (red line) scenarios. Reproduced from [Sch10] with permission.

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(The incoming spikes are on a time scale significantly slower than membrane dynamics such as w.) So this formulation of reliability is substantially delayed with respect to the dynamics of the system. This delay creates the type of results seen in figure 10.18A (plate 28) for control based on turning the stimulator on when estimated reliability is too low.

On the other hand, what is delayed for the past is predictive of the future. If you use low reliability as a control variable, by the time the moving average of reliability decreases to an arbitrary threshold, turning on stimulation will likely come too late. But for periodic fluctuations, for which Parkinson's disease abounds, you can invert your control variable. This uses *high* reliability as a control threshold. By the time reliability becomes high enough, the lack of inhibition will be drawing to a close, and the incipient upturn in inhibition will be imminent. Such results can be interesting, as shown in the improved control in figure 10.18B. In these examples attempting to use estimated reliability as a control variable, the control reliability with respect to GPi output never rose above 46 to 50%. On the other hand, our goal is to treat thalamic reliability, not the waveform of GPi output, and calculating control through running estimates of thalamic reliability is something we are presently exploring in more complete models.

Given these insights, we have begun to explore the full sparse structured Rubin and Terman model [RT04] with feedback control. In figure 10.19A, we show an individual TC cell response in the Parkinsonian state without feedback control. We now apply periodic high-frequency DBS to the STN nucleus (166 Hz), and calculate that the reliability of the TC cells improves from 45% to 100% in figure 10.19B. Now we simulate recording from GPi with a single macroscopic electrode, produce a composite summed measurement as if we were using a typical DBS electrode contact to record. We have found that abetting proportional control with a bias term, basically a proportional-integral control law strategy, we can achieve high reliability with a reduction of over 2 orders of magnitude in control energy applied, as shown in figure 10.19C, by delivering stimulation back to GPi.

To summarize this section, we have explored a variety of dynamics for controlling the thalamic cells. These examples envision direct stimulation of GPi, and our results are consistent with those of Pirini and colleagues [PRSC08], demonstrating that open loop stimulation of GPi can be a strong suppressor of thalamic activity (figure 10.13). Control through estimation of nullclines is not a trivial problem (figure 10.14), but we have not approached this with the addition of nullcline constraints and inertia that would further improve this strategy. It is interesting that direct estimation of the synaptic currents bombarding the thalamus (figure 10.17) was not as effective as estimating the T-current activation (figure 10.15) in these proportional control algorithms. Similarly, control from reliability estimates (figure 10.18) was not as effective as estimating T-current activation. One reason why T-current may have been so effective is that the time constant on the T-current activation equation is intermediate between the rapid estimation of GPi input currents and the relatively slow reliability estimates. This intermediate time constant may help damp out the relatively high

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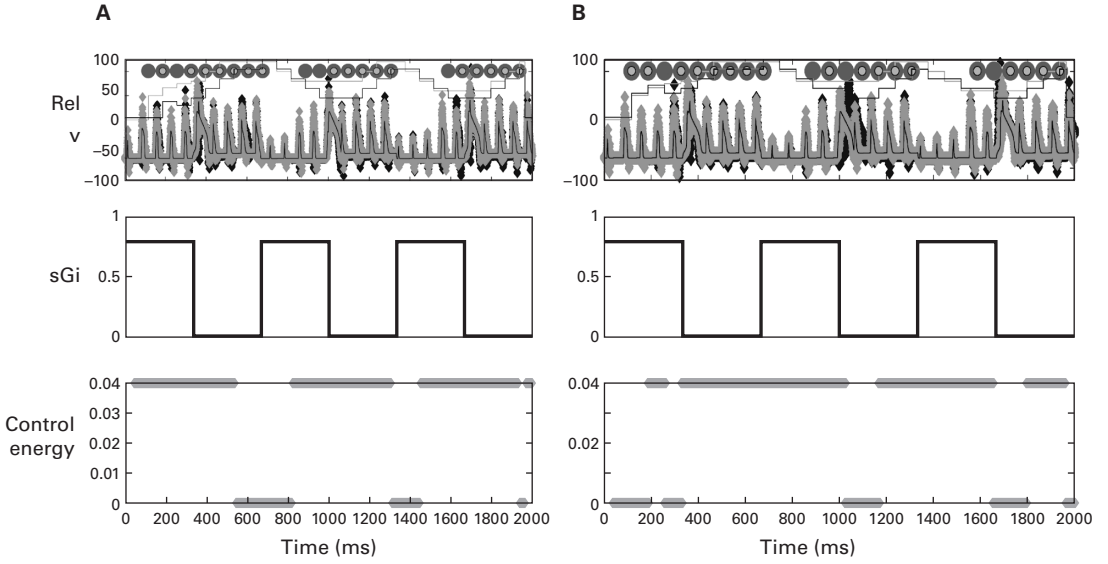


Figure 10.18 (plate 28)
Control of a TC reduced-cell model using reliability as a control parameter. A shows a threshold of turning on GPi stimulation when reliability is <0.9. B shows a different strategy, using an inverse approach. In B, control is turned on when reliability is >0.5. Note that the relevant reliability in both examples is the controlled (red) piecewise continuous line in the upper panel (the blue reliability line is the uncontrolled state shown for comparison). The inherent delays in employing the moving average of reliability can be exploited so that inverse reliability control can be more reliable than when using a more intuitive strategy based on turning on stimulation when reliability falls. Reproduced from [Sch10] with permission.

levels of noise that we have imposed on this system, but still offer suitable responsiveness for an essential component of the pathological dynamics. Whether more instantaneous estimates of reliability would serve to improve this situation is unclear (the reliability uncertainty would increase with less averaging). Nevertheless, in principle, and in full model simulations, reliability is a compelling control variable at the heart of the control problem in Parkinson's disease. Last, regardless of which method is chosen, adaptively adjusting covariance inflation and control gain (figure 10.16) is a fundamental necessity to optimize such algorithms over time.

10.11 Looking Foward

I have just touched on the potential strategies for the use of computational models in Parkinson's disease control in this chapter. Sketching out the unstudied issues is worthwhile.

The calculations in figures 10.14 through 10.18 assume that two electrodes are inserted—a recording electrode in the thalamus and a stimulating electrode in the GPi or STN. The ideal Parkinson's controller would work off of a single electrode, albeit one with multiple

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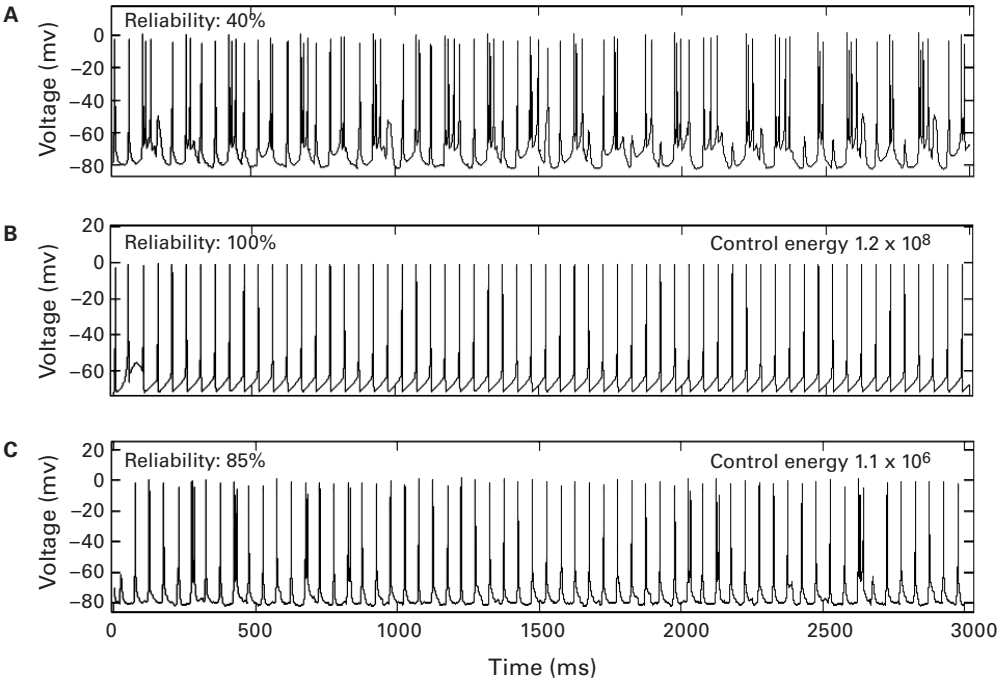


Figure 10.19
Feedback control using full model of Rubin and Terman [RT04]. A shows a TC cell in the Parkinsonian state. The reliability is 45%. B shows high-frequency DBS (166 Hz) applied to STN. The reliability of the TC cells is 100%. C shows feedback control using a proportional-integral scheme (see text for details). The reliability is 85%, in the setting of a 100-fold reduction in control energy. Figure and results are courtesy of collaboration with Patrick Gorzelic and Alok Sinha.

contacts, inserted into just one nucleus (an example is modeled in figure 10.19C). We can employ separate contacts for recording and stimulation along the electrode shaft. Perhaps picking up the oscillatory rhythms from GPi or STN would be sufficient for feedback control of those same nuclei. On the other hand, the models presented here give us the freedom to take such oscillatory dynamics from GPi or STN and estimate the reliability of the thalamus. We might never be able to record a good estimate of sensorimotor input to the actual thalamus, but we can provide such signals, or a range of such signals, to a model thalamus that is functioning as an observer model system when only GPi or STN serves as the recording site.

The reduced model of the TC cell is valuable for gaining intuition. Adding back the fast Hodgkin-Huxley sodium and potassium currents may be helpful with such models in tracking the thalamic dynamics, as might using an ensemble of such cells. But one of the appeals of such reduced models is that they can represent ensemble activity. Using a scaled-up version of this reduced TC cell, *renormalized* [Fis98] to represent an ensemble of such cells, might well be an effective way to track thalamic dynamics in real brains.

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As shown in figure 10.19, we are presently using the full Rubin-Terman model [RT04] to explore set-up of a control system by working with the more complete computational Parkinsonian brain. All of the same principles mentioned previously for the reduced model tracking are applicable in the full model. But one of the interesting facets of contrasting full and reduced models is that, beyond the obvious computational overhead of computing additional variables and parameters, the need to fit these additional variables and parameters may render more complete models *less accurate* in tracking complex systems than more reduced models. This seems counterintuitive, but the use of reduced models is commonplace in meteorological forecasting in place of the full atmospheric dynamical equations [Kal03], and in recent fluid dynamics experiments, reduced models have also proven useful in such data assimilation frameworks [CHO$^+$09] (chapter 8). An important feature of such reduced models for tracking scenarios is that the optimal parameters in these inadequate models are often nonphysical [CHO$^+$09, SS09]. But what matters for Parkinson's disease patients is that we optimize our controllers and improve symptoms better than with open loop approaches, not that we reverse engineer their failing basal ganglia. Model validation is not our primary goal in neural system control.

Recent work has extended the model of Rubin and Terman [RT04] to take into account more biologically relevant connections, the *direct* versus *indirect* pathways, from striatum to the structures of the basal ganglia (figure 10.20). The further incorporation of relevant model components may well give greater fidelity to the realistic basal ganglia dynamics in Parkinson's disease. But with control as goal, such fidelity through complexity will need to be balanced against accuracy of data assimilation and control metrics. One of the important issues raised by Pirini and Colleagues [PRSC08] is that a more complex model of the thalamic cell's function, beyond the simple relay, is likely important. One example of this is *action selection* theory, which envisions that the basal ganglia serve to select from competing neuronal efforts for access to the final common path of motor movement

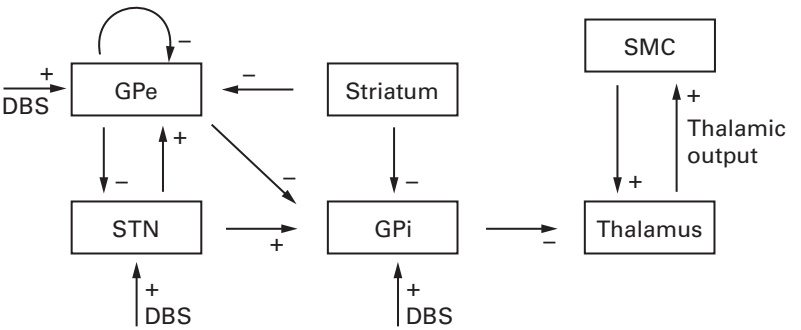


Figure 10.20
The extended Rubin-Terman model as proposed by Pirini and colleagues. Reproduced from [PRSC08] with permission.

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[HSG06]. Furthermore, as in human patient experience, the DBS target sites are not equivalent. Indeed, maintaining the flexibility to perform model-based control off of STN, GPi, or VIM, depending on a patient's symptom complex, is a challenge for future work.

Parkinson's disease will likely be the first dynamical human disease in which model-based control principles will show efficacy. The bar is not really that high. Were we merely to demonstrate that we can significantly extend the battery life of implanted stimulators, the control algorithms would be valuable. But as I hope the preceding calculations indicate, such device longevity may well be accompanied by improved performance. The parameter space that we presently face clinically in adjusting a patient's stimulator is, for all practical purposes, infinite. Adaptive algorithms that can continually optimize stimulator performance will give our present empirical trial-and-error efforts very good competition. As long as our devices operate within the electrical safety limits, and incorporate present DBS protocols as alternatives if the adaptive protocals are not better, our patients have nothing to lose and improved lives to gain.

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